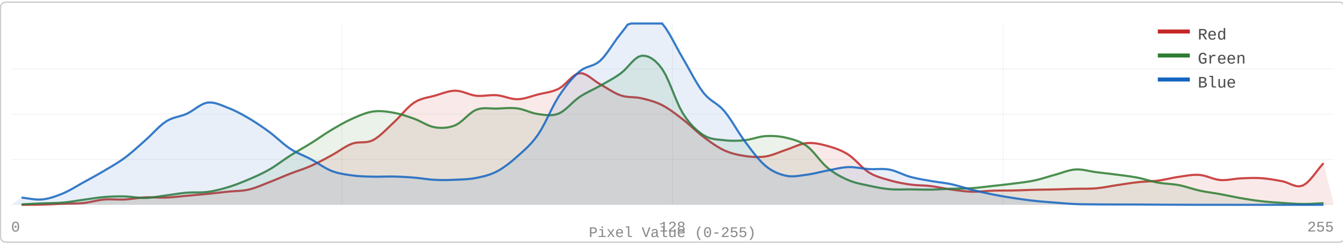
Decorrelation Gap: KODIM19

512x768 | 24-bit RGB | PNG (lossless)

Lighthouse in Maine | Strongly one-dimensional (PC1 = 93.36%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	**1.0000**	0.9668	0.8155
G	0.9668	**1.0000**	0.9151
B	0.8155	0.9151	**1.0000**

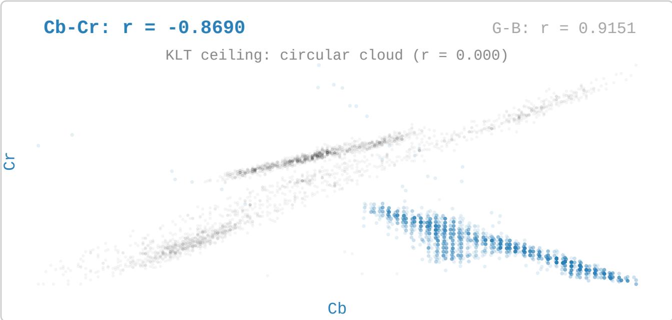
Cb-Cr: -0.869

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	**1.0000**	-0.2792	0.3006
Cb	-0.2792	**1.0000**	-0.8690
Cr	0.3006	-0.8690	**1.0000**

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.966830	-0.279223	0.2792
R-B → Y-Cr	0.815462	0.300638	0.3006
G-B → Cb-Cr	0.915071	-0.869033	0.8690
Avg |r|	0.899121	0.482965	0.4830

Decorrelation Gap: KODIM19

512x768 | 24-bit RGB | PNG (lossless)

Lighthouse in Maine | Strongly one-dimensional (PC1 = 93.36%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

46.3% Moderate

0% BT.601 leaves all 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

13.0%

0% Fully predictable 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	46.14 dB
G	49.91 dB
B	43.80 dB
Overall	45.95 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6208	-0.5770	-0.5307	6370.58
PC2	0.6120	0.0664	-0.7881	428.12
PC3	-0.4900	0.8140	-0.3119	24.66

8. PCA Derived Metrics

Condition Number

258.32

PC1/PC2: 14.88 | PC2/PC3: 17.36

Blue Channel Independence

13.0%

Variance not captured by PC1

PC1 Dominant Channel

R

Loading: R (0.62)

9. Redundancy Profile

Classification: Strongly one-dimensional | CN: 258.32 | PC1: 93.4%

RGB Correlation: Avg |r| = 0.899121 | R-G: 0.9668 | R-B: 0.8155 | G-B: 0.9151

YCbCr Residual: Avg |r| = 0.482965 | Y-Cb: -0.2792 | Y-Cr: 0.3006 | Cb-Cr: -0.8690

Efficiency: 46.3% of KLT ceiling | Gap: 0.4830 avg |r| remaining

Blue Channel: 13.0% independent | 4:2:0 PSNR: 45.95 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"

[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.

[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.

[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.